

Topic: Stack

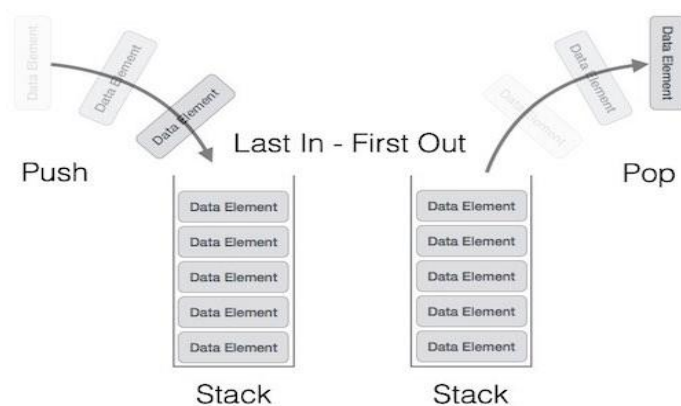
Course: CSE 207 Data Structures
Instructor: Ahmed Abdal Shafi Rasel

Objective: The objective of this lab is to provide basic concept of stack. At the end of the lab, students are able:

- To learn how to create a stack
- To learn how to perform push and pop operation in stack
- To learn how to use stack for parsing unmatched parenthesis in an algebraic expression
- To learn how to use stack for reversing data.

Lab Activities:

A **stack** is an Abstract Data Type (ADT), commonly used in most programming languages. It is named stack as it behaves like a real-world stack, for example – a deck of cards or a pile of plates, etc. A real-world stack allows operations at one end only. This feature makes it LIFO data structure. In stack terminology, insertion operation is called PUSH operation and removal operation is called POP operation.



Now you have to perform the following lab task on stack. At first build a stack ADT. Use the way that I have taught you in class. Then use the stack ADT in a new program file and implement the exercises mentioned in the lab manual

Problems to Solve:

Task 1	Implement a stack using an array. • Implement push(), pop(), peek(), and isEmpty() functions. • Ensure proper handling of stack overflow and stack underflow.
Task 2	Implement a stack using a linked list. • Implement push(), pop(), peek(), and isEmpty() functions. • Compare with array-based implementation in terms of efficiency.
Task 3	Reverse a string using a stack. • Input: "HELLO" • Output: "OLLEH"
Task 4	Write the program called copyStack that copies the contents of one stack into another. The algorithm passes two stacks, the source stack and the destination stack. The order of the stacks must be identical.
Task 5	Check if a given string has balanced parentheses. • Input: "{[(())]}" → Output: "Balanced" • Input: "{[(])}" → Output: "Not Balanced"
Task 6	Evaluate a postfix expression using a stack. • Input: "5 3 + 8 2 - *" • Output: 40 • (Perform operations in the correct order using a stack)